



Reliability-based robust geotechnical design of spread foundations using multi-objective genetic algorithm

C. Hsein Juang^{a,b,*}, Lei Wang^a

^a Glenn Department of Civil Engineering, Clemson University, Clemson, SC 29634, USA

^b National Central University, Jhongli City, Taoyuan County 32001, Taiwan, ROC

ARTICLE INFO

Article history:

Received 18 April 2012

Received in revised form 29 September 2012

Accepted 7 October 2012

Available online 12 December 2012

Keywords:

Reliability

Uncertainty

Failure probability

Limit states

Optimization

Robust design

Spread foundations

ABSTRACT

This paper presents a new geotechnical design concept, called robust geotechnical design (RGD). The new design methodology seeks to achieve a certain level of design robustness, in addition to meeting safety and cost requirements. Here, a design is considered *robust* if the variation in the system response is insensitive to the variation of noise factors such as uncertain soil parameters and construction quality. When multiple objectives are considered, a single best design may not exist, and a trade-off may be necessary. In such a case, a genetic algorithm is adopted for multi-objective optimization and a Pareto Front, which describes a trade-off relationship between cost and robustness at a given safety level, is established. The new design methodology is illustrated with an example of spread foundation design. The significance of the RGD methodology is demonstrated.

© 2012 Elsevier Ltd. All rights reserved.

1. Introduction

It is well recognized that uncertainty of soil parameters is usually unavoidable in the geotechnical design [1]. The uncertainty in the soil parameters, as well as the uncertainty in the adopted solution model, can lead to the uncertainty in the solution (e.g., predicted response or performance of a system). In a deterministic approach, the engineer uses factors of safety that have been “calibrated” by experience to cope with the uncertainty in the solution (i.e., predicted response). Of course, the factor of safety adopted in a particular design depends not only on the degree of uncertainties but also on the consequence of failure; in other words, it depends on the “calculated risk” [2,3]. To better deal with the uncertainties, the probabilistic or reliability-based approach that considers explicitly the uncertainties in the soil parameters and solution model has been proposed (e.g., [3–12]).

In a traditional geotechnical design, regardless of whether the deterministic approach or the probabilistic approach is adopted, the design is often based on a trial-and-error process considering safety and cost. Safety is usually checked first to ensure the candidate design satisfying the prescribed “safety” requirements (in terms of factor of safety or probability of failure). Then the design

with the least cost is selected from the pool of all acceptable designs that have been screened based on safety requirements [13–15]. Thus, the reliability-based design is quite straightforward if the results of the reliability analysis are accurate and precise so that there will be no question whether a given design satisfies the safety requirement. The accuracy and precision of a reliability analysis, however, depends on how well the random soil parameters are characterized. If the knowledge of the statistical distribution of soil parameters is “perfect”, the results of reliability analysis will be accurate and precise and the reliability-based design can be easily implemented with least cost objective constrained with a minimum reliability index requirement.

In a real-world geotechnical project, the distribution of a soil parameter is quite uncertain due to lack of data, measurement error, and/or error caused by use of empirical correlations. The variation range of geotechnical parameters is usually quite large [16] and thus the variation can be either overestimated or underestimated. Such overestimation or underestimation of the variation of soil parameters can lead to over-design or under-design.

While reduction of the uncertainty in soil parameters is important, which should be pursued whenever it is deemed cost-effective, in this paper, we focus on a different approach by achieving robustness in the design without eliminating the sources of uncertainty. Here, a design is considered *robust* if the variation in the system response is insensitive to (or robust against) the variation of uncertain soil parameters (called noise factors in this paper). The essence of a robust design is to select a design (through

* Corresponding author at: Glenn Department of Civil Engineering, Clemson University, Clemson, SC 29634, USA.

E-mail address: hsein@clemson.edu (C.H. Juang).

the adoption of a set of design parameters) that yields a minimal variation in the system response without eliminating the sources of uncertainty or reducing the level of uncertainty.

In this paper, a robust geotechnical design (RGD) methodology is proposed to fulfill the goal of minimizing the effects of the uncertainty of soil parameters. Robust design was originated from the field of Quality Engineering [17], which has been applied to many structural or mechanical problems in the last two decades [18–22]. One widely accepted definition of robust design [18] is manipulating design parameters (i.e., the so-called “easy to control” factors) so that the system response of the design is insensitive to, or robust against, the variation of noise factors (i.e., the so-called “hard to control” factors). In a geotechnical design, the noise factors are the uncertain soil parameters and other factors such as those related to construction. Thus, in a robust design, regions in the design space that yield low variation in the system response should be sought. The robust design will have an acceptable performance even with unexpected variation in soil parameters.

In this paper, the design of spread foundation is employed as an example to illustrate the robust design philosophy. In the sections that follow, a brief review of a reliability-based model for design of spread foundation [14] is first provided. Then, the RGD framework is presented with an illustrated example. It should be noted that the RGD methodology presented in this paper does not depend on the reliability-based procedure described by Wang [14]; rather, the latter is used for convenience in presenting a reliability-based RGD methodology.

It should be noted that robust geotechnical design (RGD) is not a design methodology to compete with the traditional approach such as the reliability-based design; rather, it is a design strategy to complement the traditional methods. With the RGD approach, the focus is to satisfy three design requirements, namely safety, cost, and robustness (against the variation in system response caused by noise factors). The safety requirement is usually implemented through constraints of reliability, and hence the design becomes a bi-objective optimization problem. For the bi-objective problem examined in this paper, it is possible that no single best solution exists that is most optimal with respect to both objectives (cost and robustness). In such a situation, a detailed study of the trade-offs between these objectives can lead to a more informed design decision.

2. Reliability-based design of spread foundation

A review of reliability based design of a spread foundation [14] based on the design example presented in an international workshop of Eurocode 7 [23,24] is provided herein. As illustrated in Fig. 1, a spread foundation is to be installed in a stiff till with a deterministic total unit weight of $\gamma = 22 \text{ kN/m}^3$ and mean effective friction angle ϕ' of 36.4° ($c' = 0$). The spread foundation is designed to support a deterministic permanent load $G = 900 \text{ kN}$ and a random variable load component of Q . The weight of foundation is considered herein by assuming a unit weight of foundation (concrete) as $\gamma_c = 24 \text{ kN/m}^3$. The foundation is founded at just above the ground water table with a foundation depth of $D = 0.8 \text{ m}$. Other parameters regarding soil and loading properties are listed in Table 1. It should be noted that the assumed coefficient of variation (COV) of the coefficient of volume compressibility m_v is slightly different from the one assumed by Wang [14]. The typical range of COV of m_v is estimated by Orr and Breyssse [24] to be 0.2–0.4, and thus a mean value of 0.3 for COV of m_v is selected in this example.

In Wang [14], the spread foundation is assumed to be square foundation and only footing width B is modeled as a design parameter. Furthermore, in Wang [14], the footing width B is assumed to

vary from a minimum value of 1.0 m to a maximum value of 3.0 m with an increment of 0.1 m and thus, only 21 possible designs exist in the design pool. In this paper, both the footing width B and footing length L are considered as design parameters; and B and L are further assumed to vary from a minimum value of 1.0 m to a maximum value of 4.0 m with an increment of 0.1 m. Thus, the number of possible designs in the design pool is 961. For a typical rectangular footing, the range of the length-to-width ratio (L/B) is between 1 and 10 [25]. When screened with this geometry constraint ($1 \leq L/B \leq 10$), the design pool is reduced to 496 designs.

The procedure for calculating the ultimate limit state (ULS) and serviceability limit state (SLS) capacity of spread foundation proposed by Orr [23] and Orr and Breyssse [24] is adopted herein for reliability-based design. The drained ULS capacity (R_{uls}) of spread foundation with a width B , length L , and Depth D can be calculated based on Annex D of Eurocode 7 [14,26] as follows:

$$R_{uls} = [(1/2)B(\gamma - \gamma_w)N_\gamma s_\gamma + \gamma DN_q s_q](BL) \quad (1)$$

where γ = unit weight of soil; γ_w = unit weight of water; N_γ and N_q are bearing capacity factors and s_γ and s_q are shape factors, defined as follows:

$$N_\gamma = 2(N_q - 1) \tan \phi' \quad (2)$$

$$N_q = e^{\pi \tan \phi'} \tan^2(45 + \phi'/2) \quad (3)$$

$$s_\gamma = 1 - 0.3(B/L) \quad (4)$$

$$s_q = 1 + (B/L) \sin \phi' \quad (5)$$

The ULS failure is checked by comparing the bearing capacity (R_{uls} , as “resistance”) with the applied loading $G + Q$. The condition, $R_{uls} < G + Q$, denotes ULS failure of the spread foundation. The SLS failure is checked by comparing the estimated total foundation settlement s_t with the maximum allowable settlement of 25 mm [24]. The total foundation settlement is constituted by two parts, the immediate settlement s_i and the consolidation settlement s_c . Determination of s_i is based on the equations given in Annex F.2 of Eurocode 7 [14,26]:

$$s_i = \frac{p(1 - \nu_u^2)Bf}{E_u} \quad (6)$$

where p is the net bearing pressure at the base level of the foundation; ν_u is the undrained Poisson's ratio ($\nu_u = 0.5$); f is the settlement coefficient ($f = 1.12$ in this example as per Orr and Farrell [26]), E_u is the undrained Young's modulus for till, which is calculated as follows:

$$E_u = 750c_u \quad (7)$$

The consolidation settlement is determined by dividing the ground below the foundation into k layers of equal thickness and summing the settlement of each layer using the following equation [24]:

$$s_c = \mu \sum_{i=1}^k m_{vi} h_i \Delta \sigma'_i = \mu m_v h \sum_{i=1}^k \alpha_i p \quad (8)$$

where μ is the settlement reduction coefficient accounting for the fact that consolidation is not one-dimensional, which is determined as 0.55 for this example [27]; m_v is the coefficient of volume compressibility ($m_v = 0.0015 \text{ m}^2/\text{MN}$); h is the thickness of each layer; $\Delta \sigma'_i$ is the increase of vertical effective in i th layer determined by $\alpha_i p$; α_i is the coefficient for increase of vertical effective stress in i th layer determined using following equation [28]:

$$\alpha_i = \frac{2}{\pi} \left[\tan^{-1} \frac{BL}{4z_i R_3} + \frac{BLz_i}{4R_3} \left(\frac{1}{R_1^2} + \frac{1}{R_2^2} \right) \right] \quad (9)$$

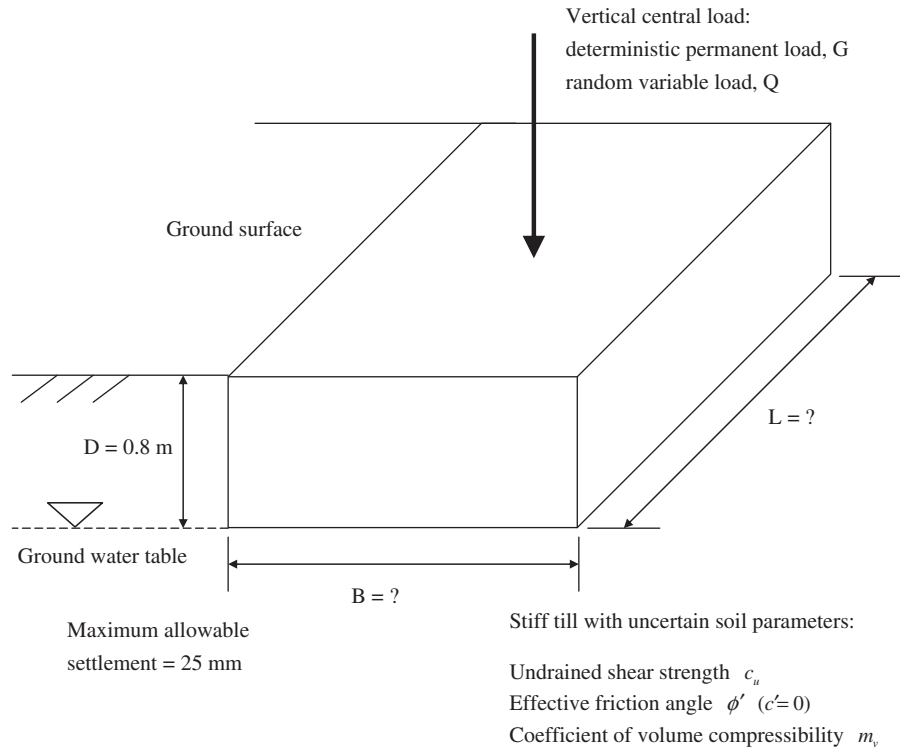


Fig. 1. A spread foundation design example (modified after [14]).

Table 1
Statistics of uncertain variables in the design example (modified after [14]).

Uncertain variables	Mean	Coefficient of variation (COV)
G	900 kN	0
Q	458.7 kN	0.15
ϕ'	36.4°	0.08
c_u	235.3 kPa	0.30
m_v	0.01875 m ² /MN	0.30

Table 2
Summary of unit price for spread foundation (source data from [29]).

Work task	Unit	US national average unit price (USD)
Excavation	m ³	25.16
Formwork	m ³	51.97
Reinforcement	kg	2.16
Concrete	m ³	173.96
Compacted backfill	m ³	3.97

where z_i is the depth of i th layer, and R_1, R_2 and R_3 are determined using the following equations [28]:

$$R_1 = \sqrt{\frac{B^2}{4} + z_i^2} \quad (10)$$

$$R_2 = \sqrt{\frac{L^2}{4} + z_i^2} \quad (11)$$

$$R_3 = \sqrt{\frac{B^2}{4} + \frac{L^2}{4} + z_i^2} \quad (12)$$

It should be noted that for SLS capacity of the foundation, the settlement coefficients in Eqs. (6) and (8), $f = 1.12$ and $\mu = 0.55$, are suitable only for design of flexible rectangular foundations in stiff soils, as in this example. For the foundation design with a different foundation shape, foundation depth, foundation stiffness or soil stiffness, different values for f and μ may be more suitable. The reader is referred to [26,27] for proper selection of these values.

For a given set of B, L and D , if the bearing capacity (R_{uls}) is smaller than the loading $G + Q$, the ULS failure is said to occur. If the calculated total settlement s_t is greater than 25 mm, the SLS failure is said to occur. Reliability-based or probability-based design can be implemented by specifying a target failure probability evaluated based on these ULS and SLS failure definitions.

3. Estimation of cost for spread foundations

The total cost for spread foundation construction is estimated based on Wang and Kulhawy [29] using following equation:

$$Z = Q_e c_e + Q_f c_f + Q_c c_c + Q_r c_r + Q_b c_b \quad (13)$$

where Q_e, Q_f, Q_c, Q_r, Q_b = quantity for five items of foundation construction work including excavation, formwork, concrete, reinforcement and compacted backfill, respectively; c_e, c_f, c_c, c_r, c_b = unit price for excavation, formwork, concrete, reinforcement and compacted backfill respectively. Table 2 summaries the US average unit price for construction of spread foundation [29]. The determination of quantity for five items Q_e, Q_f, Q_c, Q_r, Q_b of foundation construction is based on the design variables B, D and L . The reader is referred to Wang and Kulhawy [29] for details. When D is a fixed value as in the present example, the total cost for spread foundation is a function of design parameters B and L .

4. Methodology for reliability-based robust geotechnical design

In the RGD scheme, the robustness is achieved by minimizing the standard deviation of the system response (in the context of this study, calculated failure probability) while meeting the conventional target reliability requirement. Although the robustness may be interpreted differently (e.g., [17,30–34]), in this paper a

geotechnical design is considered *robust* if the standard deviation of the system response is small. This interpretation of robustness is also seen in mechanical and structural design optimization [20,21,35,36].

It is noted that the probability of failure (p_f^{ULS} based on ULS or p_f^{SLS} based on SLS) is usually determined using Monte Carlo Simulation (MCS) or reliability methods such as first order reliability method (FORM) that requires knowledge of the variation of soil parameters. If the actual variation of soil parameters is underestimated, the design obtained based on reliability (i.e., failure probability) and least-cost consideration may violate the safety requirement (which is expressed in the form of $p_f^{ULS} < p_T^{ULS}$ where p_T^{ULS} is the target failure probability). One method to avoid this problem is to consider robustness explicitly in the design so that the system response (or failure probability here) is insensitive to the variation in noise factors (i.e., uncertain soil parameters here). Thus, the essence of the robust geotechnical design (RGD) is to ensure the robustness against the variation in noise factors by manipulating the design parameters (B and L of spread foundation here). In other words, the effects of the variation of soil parameters are reduced by adjusting design parameters to achieve robustness, rather than eliminating the sources of uncertainty or reducing the level of uncertainty.

Of course, the design must also be cost-efficient. Thus, cost has to be a design objective in addition to robustness. Thus, the robust geotechnical design scheme may involve multi-objective optimization. In this paper, the multi-objective genetic algorithm optimization technique is employed to implement the RGD, considering safety, robustness, and cost simultaneously in the design.

4.1. Estimation of the coefficients of variation of soil parameters

Because sample size (i.e., number of specimens tested) is usually small in geotechnical practice (due mostly to the budget constraint), the statistics of soil parameters derived from a few testing results may not reflect the statistics of the real soil properties. Generally, the “population” mean can be adequately estimated from the sample mean even when the sample size is quite small [6]. Thus, in this paper, the mean of each noise factor (uncertain soil parameter) is considered as fixed value. However, the estimation of standard deviation of “population” may be subjected to error when the sample size is small. With insufficient data, the variation of soil parameters is usually estimated based on the prior knowledge in the form of local or regional experience of particular soil condition or published literature [24]. In fact, Orr and Breyse [24] provided typical ranges for the coefficient of variation (COV) of three soil parameters ϕ' , c_u and m_v for the design example of spread foundation. The COV of ϕ' , denoted as $COV[\phi']$, typically ranges from 4% to 11%; the COV of c_u , denoted as $COV[c_u]$, typically ranges from 20% to 40%; and the COV of m_v , denoted as $COV[m_v]$, typically ranges from 20% to 40%. Thus, the COVs of ϕ' , c_u and m_v should be considered as random variables.

As an example to illustrate the robustness concept presented herein, the mean of $COV[\phi']$, denoted as $\mu_{COV[\phi']}$, is assumed to be 0.08, and the coefficient of variation of $COV[\phi']$, denoted herein as $\delta_{COV[\phi']}$, is assumed to be 25% (roughly to cover the typical range of $COV[\phi']$). Similarly, the mean of $COV[c_u]$, denoted as $\mu_{COV[c_u]}$, is assumed to be 0.30, and the coefficient of variation of $COV[c_u]$, denoted herein as $\delta_{COV[c_u]}$, is assumed to be 17% (roughly to cover the typical range of $COV[c_u]$); the mean of $COV[m_v]$, denoted as $\mu_{COV[m_v]}$, is assumed to be 0.30, and the coefficient of variation of $COV[m_v]$, denoted herein as $\delta_{COV[m_v]}$, is assumed to be 17% (roughly to cover the typical range of $COV[m_v]$). Furthermore, all random variables used in the example are assumed to follow lognormal distribution as recommended by Wang [14], and these random variables are further assumed independent from each other [14].

As is shown later, the robustness of a reliability-based design is achieved if the probability of failure of a design is insensitive to the variation of the estimated COVs of the input soil parameters. For a given set of fixed COVs of soil parameters (ϕ' , c_u and m_v in this example), the failure probability determined with a reliability analysis such as FORM [4,37] will be a fixed value. If the COVs of these soil parameters are treated as random variables, as described previously, then the resulting failure probability will depend on the actual distribution employed to characterize the uncertainty of such COVs. Thus, the variation of the failure probability has to be evaluated. In this paper, this variation is evaluated using point estimate method (PEM).

4.2. PEM procedure for deriving mean and standard deviation

FORM procedure for computing the system response (either p_f^{ULS} or p_f^{SLS}) can be thought of as a complex function $y = f(x)$, where y is the system response and x is a set of input variables.

The seven-point PEM procedure by Zhao and Ono [38,39] is adopted herein to compute the mean and standard deviation of both p_f^{ULS} and p_f^{SLS} for a given spread foundation design. By evaluating the deterministic models presented previously, it is found that p_f^{ULS} is affected only by noise factor ϕ' and p_f^{SLS} is affected only by noise factors c_u and m_v . The procedure of computing the mean and standard deviation of p_f^{ULS} is illustrated here as an example.

PEM uses a weighted sum of the function evaluated at a finite number of “estimating” points, which are chosen to satisfy the equivalency in moments:

$$\sum_{j=1}^m P_j (x_j - \mu_x)^k = M_{kx} \quad (14)$$

where $x_{jj} = 1, \dots, m$ are the estimating points; P_j , $j = 1, \dots, m$ are the corresponding weights; μ_x is the mean of x , and M_{kx} is the k th central moment of x .

In the modified PEM outlined by Zhao and Ono [38], the estimating points are obtained in the standard normal space. Thus, non-standard normal variables (x_j) must be transformed into standard normal variables (u_j), which can be obtained through Rosenblatt transformation [40]. The seven estimating points and their corresponding weights in the standard normal space are [38]:

$$\begin{aligned} u_1 &= 0; & P_1 &= 16/35 \\ u_2 &= -u_3 = 1.1544054; & P_2 &= P_3 = 0.2401233 \\ u_4 &= -u_5 = 2.3667594; & P_4 &= P_5 = 3.07571 \times 10^{-2} \\ u_6 &= -u_7 = 3.7504397; & P_6 &= P_7 = 5.48269 \times 10^{-4} \end{aligned} \quad (15)$$

The k th central moment of the function $y = f(x)$ in the original space can be calculated as follows:

$$\mu_y = \sum_{j=1}^m P_j f[T^{-1}(u_j)] \quad (16)$$

$$M_{ky} = \sum_{j=1}^m P_j \{f[T^{-1}(u_j)] - \mu_y\}^k \quad (17)$$

where the term $T^{-1}(u_j)$ is the inverse Rosenblatt transformation, μ_y is the mean of y , and M_{ky} is the k th central moment of y . To evaluate the variation of the ULS probability failure (p_f^{ULS}) caused by the variation of $COV[\phi']$, let p_f^{ULS} be y and $COV[\phi']$ be x . The seven $COV[\phi']$ values in the original space (x_j) are obtained through inverse Rosenblatt transformation based on the seven estimating points in the standard normal space (u_j). The mean and standard deviation of p_f^{ULS} can be readily computed with Eqs. (16) and (17). The mean and standard deviation of the SLS failure probability p_f^{SLS} can be computed in a similar way, although in this case, x is a vector con-

sisting of $COV[c_u]$ and $COV[m_v]$. The reader is referred to Zhao and Ono [38,39] for details of formulation with multiple input variables.

4.3. Reliability-based robust geotechnical design (RGD) approach

The framework for reliability based robust geotechnical design is illustrated using spread foundation design as an example. In reference to Fig. 2, the RGD approach that is “tuned” for spread foundation design is outlined in the following steps:

Step 1 involves classification of the design parameters and noise factors in the geotechnical system of interest. For the design of rectangular spread foundations, the design parameters are B and L (with a fixed $D = 0.8$ m in this example). The parameters B and L can be modeled in the discrete domain for construction convenience and the design space is thus finite with, say, M designs. In this example, 961 possible designs ($M = 961$) exist in the design space with different combinations of B and L before the geometry requirement is applied. The uncertain soil parameters ϕ' , c_u and m_v are considered as noise factors here. The statistics of the noise factors in this example are assumed based on the work of Orr and Breyse [24], as discussed previously.

Step 2 involves the assignment of COV values of the noise factors for each design. This is the first step in the inner loop (shown in Fig. 2) that is used to compute the variation of system response (i.e., failure probability) using PEM. As noted previously, the mean of each noise factor is treated as a fixed value, and its COV is assigned based on PEM procedure described previously.

Step 3 is a reliability analysis for each design in the design space. The deterministic model as defined previously for the ULS and SLS requirements for spread foundation can be adopted for probabilistic analysis. The probability of ULS or SLS failure for each design in the design space is determined using FORM in this paper. The results obtained from FORM are confirmed using MCS, although not shown here.

Step 4 is to repeat the reliability analysis procedure N times (note: $N = 7k$ where k is the number of noise factors) and then use the results to compute the mean and standard deviation of the probability of failure using PEM.

Step 5 deals with the outer loop shown in Fig. 2, in which the PEM procedure (Steps 2–4) is repeated for each of the M designs. The mean and standard deviation of the failure probability for each of the M designs is obtained. For the illustrative example presented later, the computation of mean and standard deviation of the ULS failure probability for each design involved only seven repeated reliability analysis using FORM, which is quite efficient. For more complicated problems (e.g., the performance function is an implicit numerical model), the proposed RGD approach can be computationally prohibitive. In such cases, the original numerical model may be approximated using a surrogate model (e.g., artificial neural network, support vector machine or response surface model), and then the mean and standard deviation of failure probabilities can be obtained using the proposed approach.

Step 6 is to select the “best” design. With the RGD approach, both cost and robustness are used as objectives, while safety is used as constraint, for multi-objective optimization. As noted previously, the design robustness is evaluated in terms of the standard deviation of the failure probability. In this paper, cost for each of the M designs is computed using the procedure by [29] described previously. Then, the multi-objective optimization is carried out using Non-dominated Sorting Genetic Algorithm version II (NSGA-II), developed by [41].

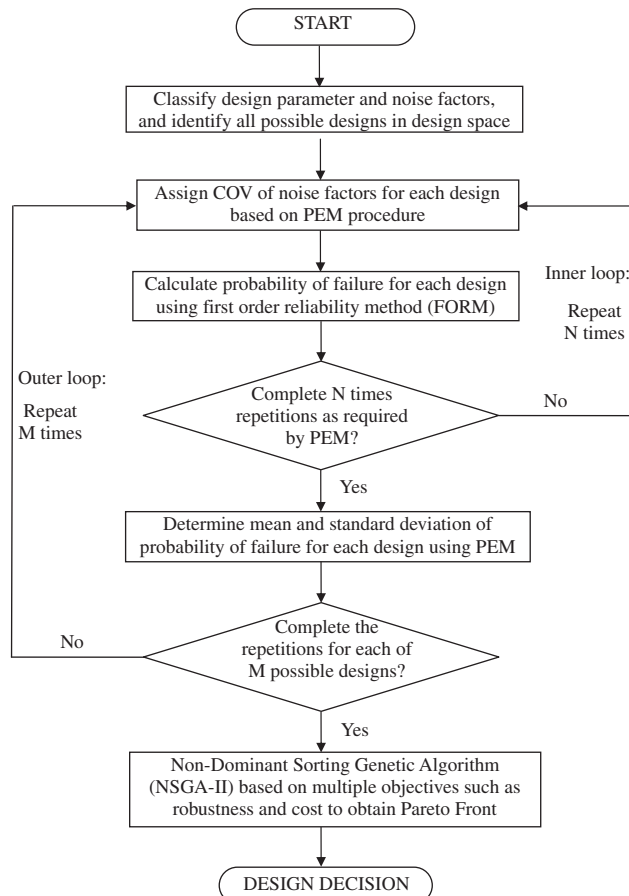


Fig. 2. Flowchart illustrating robust geotechnical design of spread foundation.

When multiple objectives are enforced, it is likely that no single best design exists that is superior to all other designs in all objectives (note: in such scenario, the imaginary point that is superior to all other designs is called *utopia* point, as shown in Fig. 3). However, a set of designs may exist that are superior to all other designs in all objectives; but within the set, none of them is superior or inferior to each other in all objectives. These designs constitute a Pareto Front. Fig. 3 shows a conceptual illustration of

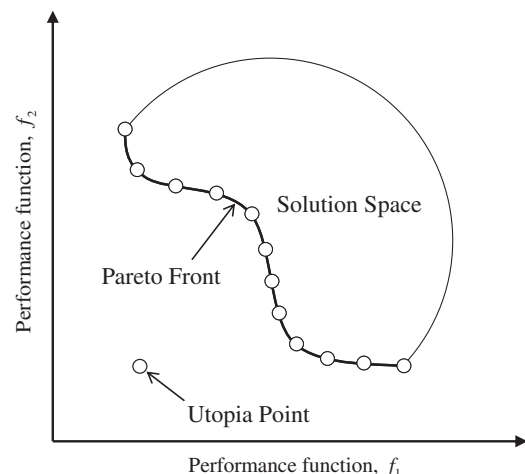


Fig. 3. Pareto Front illustrated in a bi-objective space (modified after [42]).

a Pareto Front in a bi-objective setting [42]. Each point on the Pareto Front is optimal in the sense that no improvement can be achieved in one objective without worsening in at least one other objective. Mathematically, the Pareto-Front is constructed by comparing all the points in the solution space based on the following definitions: design A is *dominated* by design B if A is inferior or equal to B in every objective measure, except one scenario that $f_i(A) = f_i(B)$ for all objectives (note: f_i is some performance function with respect to objective i where $i = 1, 2$ in this example). If one design is not dominated by any other designs, it belongs to the Pareto Front. When the optimization process yields a Pareto Front, a trade-off situation is implied.

Finally, it should be noted that the procedure described above (in reference to Fig. 2) is only one possible implementation of the RGD methodology. Other implementations may be equally effective. For example, FORM as a means to compute the failure probability for a given design with a set of known mean and COV of each of the noise factors may be replaced by MCS. Similarly, PEM as a means to compute the variation of the system response (i.e., the failure probability) may also be replaced by MCS or other means.

5. Traditional reliability-based design

The reliability based design of spread foundation without considering robustness is first presented herein to provide a reference. The spread foundation shown in Fig. 1 with pertinent data listed in Table 1 is examined. The probability of SLS and ULS failure for each design for a given vertical load consisting of a fixed permanent load component of $G = 900$ kN and a random variable load of Q is determined using FORM. This analysis is repeated for all possible designs in the design space. For illustration purpose, the results

(i.e., failure probabilities) are plotted only for selected designs with $L = 2.0$ m, 3.0 m and 4.0 m, as shown in Fig. 4.

It can be seen from Fig. 4 that the probabilities of both ULS failure and SLS failure decrease with the increase of B and L . From Fig. 4a, when L varies from 2.0 m to 4.0 m, the threshold value of B that satisfies the ULS failure probability requirement (i.e., below the dashed line) decreases from 1.8 m to 1.2 m. As can be seen from Fig. 4b, all designs in the plot satisfy the SLS failure probability requirement. The ULS failure probability requirement is more stringent than the SLS failure probability requirement and thus, in this case, the ULS controls the design of spread foundations. In fact, for all the COV levels analyzed in this study, the ULS requirement is more stringent than the SLS requirement for this shallow foundation problem. Therefore, in the subsequent analysis only the ULS failure probability is considered.

It should be noted that the spatial variation of soil properties and correlation between soil properties are not considered in this paper, and the soil properties are assumed lognormally distributed. These limitations may not be critical, as the SLS is not the controlling limit state in the present example. However, proper modeling of spatial variability and distribution of soil properties will be essential if the SLS controls the foundation design, since these factors could have a significant influence on the settlement of foundation (e.g., Jimenez and Sitar [43]).

In a reliability-based design, the reliability is generally used a constraint to screen for the acceptable designs, and then the best design is attained by selecting the least-cost design [14,15]. Although the optimal design may be selected based on the minimum life-cycle cost [44,45], in this paper only the initial cost of spread foundation is considered so that the focus can be placed on the subject of design robustness. The initial cost generally refers to the construction cost for completion of a spread foundation. In this paper, the procedure by Wang and Kulhawy [29], described previously is adopted.

The target ULS reliability index $\beta_T^{ULS} = 3.8$ (corresponding to $p_T^{ULS} = 7.2 \times 10^{-5}$) and the target SLS reliability index $\beta_T^{SLS} = 1.5$ (corresponding to $p_T^{SLS} = 6.7 \times 10^{-2}$) defined in Eurocode 7 [24] for spread foundations is adopted herein for reliability based design. If the “best” design is to be chosen based on least cost subjected to the constraints of the minimum reliability requirement ($\beta_T^{ULS} > \beta_T^{ULS} = 3.8$) and the geometry requirement ($1 \leq L/B \leq 10$), then the design with $B = 1.8$ m and $L = 2.0$ m will be selected.

To demonstrate the effect of the variation in the estimated COVs of soil parameters, two series of reliability analysis are performed. One is to determine the least cost designs for various assumed COV values (see Table 3), and the other seeks to determine the failure probability of a given design under various levels of COV (see Table 4).

From the results shown in Table 3, it is observed that the least cost designs are quite sensitive to the assumed $COV[\phi']$. On the other hand, $COV[c_u]$ and $COV[m_\nu]$ have no effects on the selected least cost designs. This is expected as the probability of ULS failure,

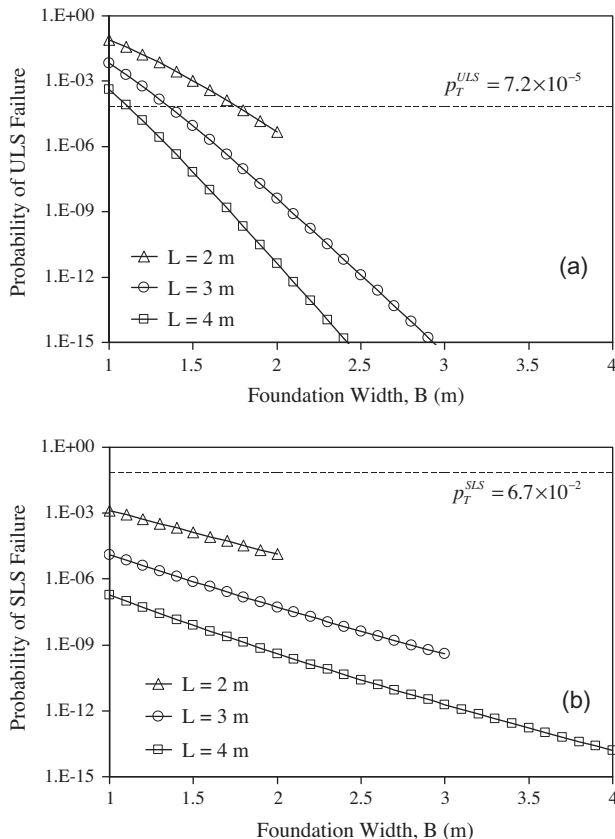


Fig. 4. Probabilities of failure of spread foundations with $COV[\phi'] = 0.08$, $COV[c_u] = 0.3$, and $COV[m_\nu] = 0.3$: (a) ULS failure and (b) SLS failure.

Table 3

Least-cost designs under various COV assumptions for soil parameters.

$COV(\phi')$	$COV(c_u)$	$COV(m_\nu)$	B (m)	L (m)	Cost (USD)
0.04	0.2	0.2	1.4	1.5	705.7
0.04	0.3	0.3	1.4	1.5	705.7
0.04	0.4	0.4	1.4	1.5	705.7
0.08	0.2	0.2	1.8	2.0	1102.7
0.08	0.3	0.3	1.8	2.0	1102.7
0.08	0.4	0.4	1.8	2.0	1102.7
0.11	0.2	0.2	2.2	2.2	1417.9
0.11	0.3	0.3	2.2	2.2	1417.9
0.11	0.4	0.4	2.2	2.2	1417.9

Table 4

ULS failure probability of a given design ($B = 1.8$ m, $L = 2.0$ m) under various COV assumptions.

COV (ϕ')	ULS failure probability, p_f^{ULS}
0.04	4.63E–14
0.05	4.51E–10
0.06	1.16E–07
0.07	4.04E–06
0.08	4.47E–05
0.09	2.44E–04
0.10	8.49E–04
0.11	2.18E–03

which controls the design of spread foundation, is only affected by $COV[\phi']$. Thus, in the next series of analysis, only variation of $COV[\phi']$ is considered.

The determination of least-cost design in a reliability-based design is meaningful only if the coefficient of variation of soil parameter (such as $COV[\phi']$) is fixed. Thus, if the COV of soil parameter is underestimated by a certain margin, then there is a chance (significant probability) that an acceptable design (a design that satisfies $\beta_p^{ULS} > \beta_T^{ULS} = 3.8$) will no longer be satisfactory. This recognition is demonstrated with results shown in Table 4, where the performance of an acceptable design ($B = 1.8$ m and $L = 2.0$ m based on an assumption of fixed $COV[\phi'] = 0.08$, and a target failure probability of $p_T^{ULS} = 7.2 \times 10^{-5}$) is reanalyzed with various levels of $COV[\phi']$. As can be seen from Table 4, if the design is implemented in a stiff till site with $COV[\phi']$ greater than 0.08, the ULS failure requirement will no longer be satisfied, as the failure probability will be greater than $p_T^{ULS} = 7.2 \times 10^{-5}$.

A review of Table 3 also indicates that even with unfavorable COV condition (for example, for the case of $COV[\phi'] = 0.11$, which is higher than the initially assumed value of 0.08), an acceptable design (for example, $B = 2.2$ m, $L = 2.2$ m) can still be found that satisfies the constraint of $p_f^{ULS} < p_T^{ULS} = 7.2 \times 10^{-5}$. In other words, some designs, albeit at higher costs, can be chosen so that they are robust against the variation in soil parameters. Of course, an overly conservative design generally violates the cost efficiency requirement.

6. Reliability-based geotechnical robust design

To investigate the significance of considering robustness in the design process, the same spread foundation problem (see Fig. 1 and Table 1) is analyzed with additional knowledge of the variation of $COV[\phi']$, $COV[c_u]$ and $COV[m_v]$. Specifically, $COV[\phi']$, $COV[c_u]$, and $COV[m_v]$ are treated herein as independent random variables with the following mean and COV values: $\mu_{COV[\phi']} = 0.08$ with $\delta_{COV[\phi']} = 0.25$; $\mu_{COV[c_u]} = 0.30$ with $\delta_{COV[c_u]} = 0.17$; and $\mu_{COV[m_v]} = 0.30$ with $\delta_{COV[m_v]} = 0.17$. Of course, these values are just adopted as an example to illustrate the concept of robustness in the design. It should be noted that only the variation in $COV[\phi']$ affects the variation of the ULS failure probability.

Following the RGD procedure described previously (in reference to Fig. 2), the mean and standard deviation of the ULS failure probability, denoted as μ_p^{ULS} and σ_p^{ULS} , respectively, can be obtained for various designs (i.e., various pairs of B and L). Fig. 5 shows the mean ULS failure probability (μ_p^{ULS}) for selected designs with $L = 2.0$ m, 3.0 m and 4.0 m. As can be seen, many designs have a mean failure probability greater than the target failure probability ($p_T^{ULS} = 7.2 \times 10^{-5}$). When all designs in the design space are screened with the constraints of both reliability (safety) requirement and geometry requirement ($1 \leq L/B \leq 10$), only 287 designs are obtained.

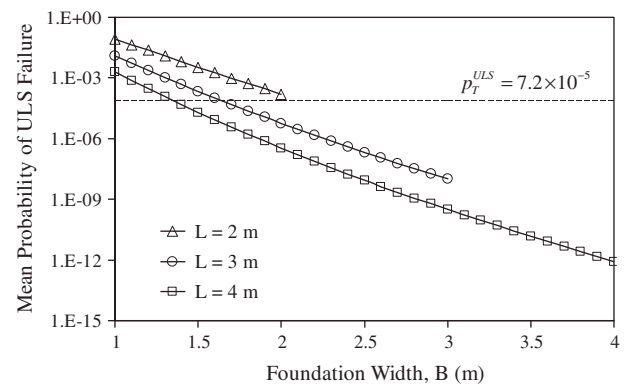


Fig. 5. Mean of the ULS failure probability for selected designs obtained using the 7-point PEM.

Fig. 6 shows the standard deviation of the ULS failure probability (σ_p^{ULS}) of each of the selected acceptable designs based on the reliability requirement. As an example, only acceptable designs with $L = 3.0$ m and 4.0 m are plotted in Fig. 6 (note that all designs with $L = 2.0$ m are unacceptable based on the failure probability requirement). As noted previously, the robustness of a reliability-based design can be gauged by the computed σ_p^{ULS} value. Thus, a design is said to have a greater robustness if σ_p^{ULS} caused by the variation of $COV[\phi']$ is smaller. If the cost of each design that met reliability and geometry requirements is plotted against the computed σ_p^{ULS} value, a trade-off relationship is observed, as shown in Fig. 7.

In Fig. 7, all data are acceptable designs based on the traditional reliability-based design approach. If the cost is the only design objective after satisfying the reliability and geometry requirements, then the design of $B = 2.1$ m and $L = 2.1$ m will be selected, which has a least cost of 1309.3 USD. However, this design has a large σ_p^{ULS} value, indicating that its robustness level is low and the design is sensitive to uncertainty in the estimated COV of soil parameters. The implication is that such a design, albeit most economical from the traditional reliability-based design viewpoint, is likely to fail the ULS failure requirement if the COV of the soil parameter is underestimated. On the other hand, if a design with a smaller σ_p^{ULS} is chosen, it will be more robust, albeit at a higher cost.

Although the trade-off relationship between cost and robustness shown in Fig. 7 is obtained without using any special technique, improved results may be obtained using a more efficient method such as multi-objective optimization using NSGA-II [41]. In the section that follows, the NSGA-II approach and the results

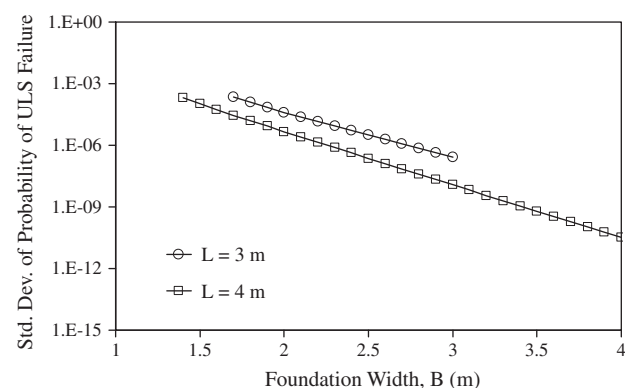


Fig. 6. Standard deviation of the ULS failure probability for selected acceptable designs using the 7-point PEM.

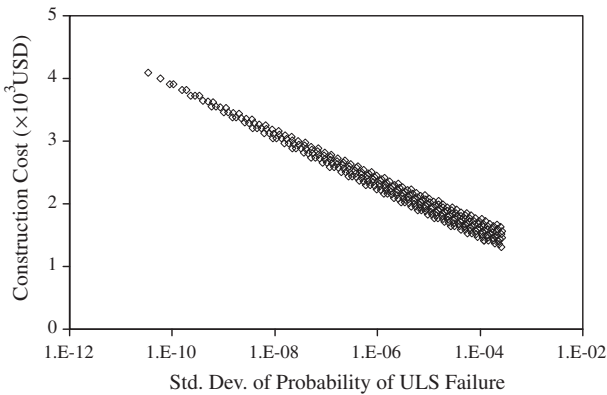


Fig. 7. Relationship between cost and standard deviation of the ULS failure probability.

obtained with this approach are presented, followed by a procedure for selecting the best designs.

7. Non-dominated sorting genetic algorithm for Pareto Front

The RGD approach considers safety, cost and robustness in the design. Safety is generally assured of through a constraint that the computed mean failure probability is less than the target failure probability. Thus, the design can be optimized with two objectives, robustness and cost, subjected to the safety constraint and any additional constraint such as geometry.

7.1. Summary of NSGA-II algorithm

In reference to Fig. 8, NSGA-II algorithm [41] is briefly described in the following. First, a random “parent population” P_0 from the design space is created. The term “parent population” is widely used in Genetic Algorithm (GA); here, it can be thought of as the first trial set of “optimal” designs that best satisfy all design objectives. Let the number of designs (or points) in P_0 be n . A series of GA operations, including selection, recombination, and mutation are performed to create the “offspring population” Q_0 of size n . Then, an iterative process is adopted to refine the parent population. In the GA terminology, each step in the iteration is a “generation”.

In the t th generation, the parent population P_t and the offspring population Q_t are combined to form a new population $R_t = P_t \cup Q_t$

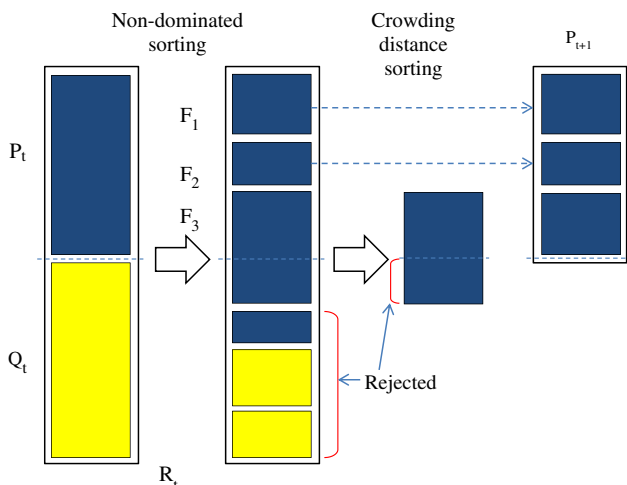


Fig. 8. An illustration of NSGA-II algorithm (modified after [41]).

with a size of $2n$. Then, the non-dominated sorting is performed on R_t , which yields different sets of points based on the non-dominated ranking. For example, the best class is labeled F_1 , and the second best class is labeled F_2 , and so on. The best n points are selected into parent population of the next generation, P_{t+1} . Using the scenario illustrated in Fig. 8 as an example, if the number of points in F_1 and F_2 is less than n , they will all be selected into P_{t+1} . Then, if the number of points in F_1 and F_2 and F_3 exceeds the population size n , the points in F_3 are sorted using the “crowding distance” sorting technique [41], which aims to maintain the diversity in the selected points. Thus, the best points in F_3 are selected to fill all remaining slots in the next population P_{t+1} [41]. After obtaining P_{t+1} in the t th generation, the algorithm proceeds to the next generation, and the process is repeated until a stable (or converged) population P_{t+1} is obtained, which is considered the optimal set of designs that are non-dominated by any other designs with respect to all objectives. The final, converged P_{t+1} is the Pareto Front.

7.2. Multi-objective optimization using NSGA-II

As noted previously, the ULS controls the design of spread foundation; thus, only the standard deviation of the ULS failure probability is integrated into the NSGA-II optimization scheme for robustness consideration. This optimization scheme for optimal design of spread foundation may be set up as follows:

Find $\mathbf{d} = [B, L]$
Subjected to: $B \in \{1.0 \text{ m}, 1.1 \text{ m}, 1.2 \text{ m}, \dots, 4 \text{ m}\}$ and $L \in \{1.0 \text{ m}, 1.1 \text{ m}, 1.2 \text{ m}, \dots, 4 \text{ m}\}$
 $\mu_p^{ULS} < p_T^{ULS} = 7.2 \times 10^{-5}$
 $\mu_p^{SLS} < p_T^{SLS} = 6.7 \times 10^{-2}$
 $1 \leq L/B \leq 10$
Objectives: Minimizing the standard deviation ULS failure probability (σ_p^{ULS})
 Minimizing the cost for spread foundation.

It should be noted that the design parameters are generated in the discrete space of the design domain using Genetic Algorithm. The population size is arbitrarily set at 100 to begin with. For this example, after 30 generations the Pareto Front has already reached a “stable” condition, meaning that little further improvement can be made. Fig. 9 shows the developed Pareto Front after reaching convergence. Note that in this example, only 80 “unique” designs are found in the final Pareto Front, compared to 287 designs that were shown in Fig. 7.

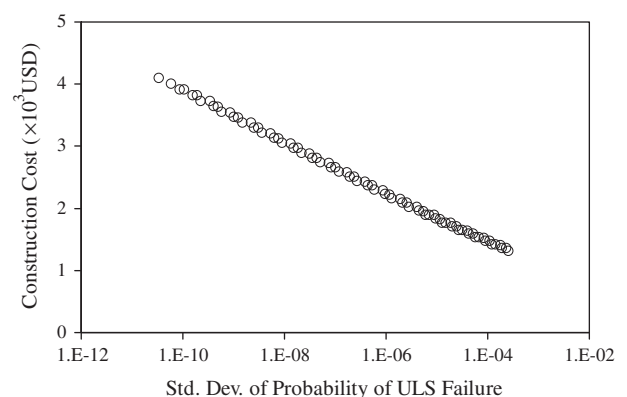


Fig. 9. The Pareto Front at convergence for spread foundation design obtained by NSGA-II based on two objectives.

Thus, the reliability-based RGD approach yields an optimal set of designs, called Pareto Front, which expresses a trade-off relationship between cost and robustness against noise factors. The designer can choose for a greater robustness (i.e., the failure probability is less sensitive to the uncertainty in the estimated variation of soil parameters) at a higher cost. If an acceptable cost range is identified (i.e., the cost must be less than some desired amount), the designer can select the most robust design from the Pareto Front. On the other hand, if a certain minimum level of robustness is desired, the design with least cost will be the best choice. To refine further the decision making process, a term called feasibility robustness is defined, which is described next.

7.3. Selection of best design based on feasibility robustness

The Pareto Front specifies a trade-off relationship between cost and robustness. While this trade-off relationship is itself a decision-making aid, further refinement may be made. Here, the concept of “feasibility robustness” [46] is introduced. For design of spread foundations, the ultimate limit state (ULS) requirement controls the design, as noted previously. The feasibility robustness is defined through the probability of meeting the ULS failure requirement: $p_f^{ULS} \leq p_r^{ULS} = 7.2 \times 10^{-5}$. It may be considered as a constraint, expressed as follows:

$$\Pr[(p_f^{ULS} - p_r^{ULS}) < 0] \geq P_0 \quad (18)$$

where $\Pr[(p_f^{ULS} - p_r^{ULS}) < 0]$ is the confidence probability that the ULS failure requirement can be satisfied, and P_0 is an acceptable level of the confidence probability selected by the designer.

Determination of the confidence probability $\Pr[(p_f^{ULS} - p_r^{ULS}) < 0]$ requires the knowledge of the distribution of p_f^{ULS} , which is generally difficult to ascertain. The histogram of the reliability index β^{ULS} (corresponding to p_f^{ULS}) of a given design (for example, $B = 2.2$ m, $L = 2.2$ m) can be obtained from Monte Carlo simulations as shown in Fig. 10, which can be approximated with a lognormal distribution. Thus, an equivalent counterpart in terms of $\Pr[(\beta^{ULS} - \beta_r^{ULS}) > 0]$, where $\beta_r^{ULS} = 3.8$ (corresponding to $p_r^{ULS} = 7.2 \times 10^{-5}$), may be used to assess the feasibility robustness.

The mean and standard deviation of β^{ULS} , denoted as μ_β and σ_β respectively, can be obtained using FORM integrated with PEM. When assuming β^{ULS} follows lognormal distribution, the confidence probability can be computed using the first order second moment method (FOSM) as follows:

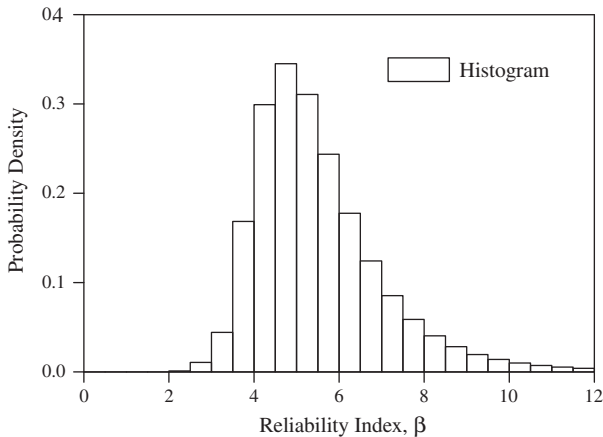


Fig. 10. Distribution of reliability index of a given design ($B = 2.2$ m, $L = 2.2$ m).

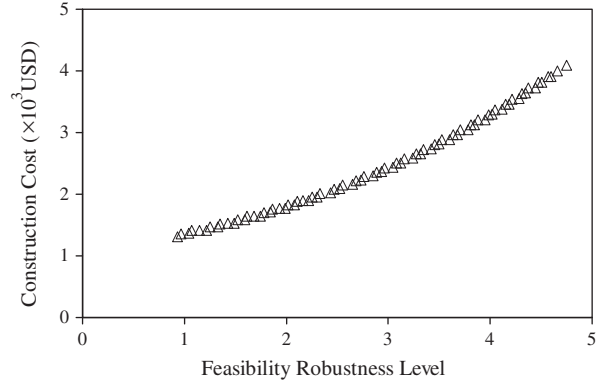


Fig. 11. Cost versus feasibility robustness for all designs on the Pareto Front.

$$\Pr[(\beta^{ULS} - 3.8) > 0] = \Phi(\beta_\beta) \quad (19)$$

where Φ is the cumulative standard normal distribution function, and β_β is defined as:

$$\beta_\beta = \frac{\ln \left[\mu_\beta / \sqrt{1 + \left(\frac{\sigma_\beta}{\mu_\beta} \right)^2} \right] - \ln(3.8)}{\sqrt{\ln \left[1 + \left(\frac{\sigma_\beta}{\mu_\beta} \right)^2 \right]}} \quad (20)$$

Thus, the β_β value may be thought of as an index for feasibility robustness. The relationship between β_β value and the cost for the 80 designs in the Pareto Front is shown in Fig. 11. As expected, the results show that a design with higher feasibility robustness costs more. By selecting a desired feasibility robustness level (in terms of β_β), the least-cost design among those on the Pareto Front can readily be determined. For example, when the feasibility robustness level is set at $\beta_\beta = 1$, which corresponds to a confidence probability of 84.13%, the least-cost design is $B = 2.1$ m and $L = 2.2$ m. The cost for this design is 1362.5 USD. When the feasibility robustness level is set at $\beta_\beta = 2$, the least-cost design is $B = 2.4$ m and $L = 2.7$ m, which corresponds to a confidence probability of 97.72%. The cost for this design that meets the feasibility robustness level of $\beta_\beta = 2$ is 1826.3 USD. Thus, the feasibility robustness offers an easier-to-use quantitative measure for making an informed design decision considering cost and robustness after satisfying the safety requirement.

8. Further considerations

8.1. Three-objective optimization considering various distribution types

In addition to the uncertainty in the sample statistics, the distribution type of the noise factors may also affects the result of reliability-based design [14]. Thus, it is desirable to identify a robust design that is insensitive to the assumed distribution. For this exercise, both normal and lognormal distributions are considered here. The mean and standard deviation of the probabilities of ULS and SLS failure based on normal distribution assumption are denoted as $\mu_{p^{ULS}}^{ULS}$ and $\sigma_{p^{ULS}}^{ULS}$, and $\mu_{p^{SLS}}^{SLS}$ and $\sigma_{p^{SLS}}^{SLS}$, respectively. Similarly, the mean and standard deviation of probabilities of ULS and SLS failure for lognormal distribution assumption are denoted as $\mu_{p^{ULS}}^{ULS}$ and $\sigma_{p^{ULS}}^{ULS}$, and $\mu_{p^{SLS}}^{SLS}$ and $\sigma_{p^{SLS}}^{SLS}$, respectively. The effect of different assumed distributions may be reduced by minimizing cost, $\sigma_{p^{ULS}}^{ULS}$, and $\sigma_{p^{SLS}}^{SLS}$ simultaneously in the NSGA-II optimization framework. Symbolically, the optimization may be formulated as follows:

Find $\mathbf{d} = [B, L]$
Subjected to: $B \in \{1.0 \text{ m}, 1.1 \text{ m}, \dots, 4 \text{ m}\}$
 $L \in \{1.0 \text{ m}, 1.1 \text{ m}, \dots, 4 \text{ m}\}$
 $\mu_{p_N}^{ULS} < p_T^{ULS} = 7.2 \times 10^{-5}$
 $\mu_{p_N}^{SLS} < p_T^{SLS} = 6.7 \times 10^{-2}$
 $\mu_{p_{LN}}^{ULS} < p_T^{ULS} = 7.2 \times 10^{-5}$
 $\mu_{p_{LN}}^{SLS} < p_T^{SLS} = 6.7 \times 10^{-2}$
 $1 \leq L/B \leq 10$
Objectives: Minimizing the standard deviation $\sigma_{p_N}^{ULS}$
Minimizing the standard deviation $\sigma_{p_{LN}}^{ULS}$.
Minimizing the cost of the spread foundation.

Fig. 12 shows the results of the three-objective optimization using NSGA-II. In the new Pareto Front that is optimized for three objectives, 69 designs are found, compared to 80 designs in the previous optimization analysis based on two objectives. The results show that $\sigma_{p_N}^{ULS}$ generally increases with the increase of $\sigma_{p_{LN}}^{ULS}$, and both $\sigma_{p_N}^{ULS}$ and $\sigma_{p_{LN}}^{ULS}$ have a trade-off relationship with the cost (although not easy to see in this 3-D plot). Fig. 13 shows the new Pareto Front plotted in the same 2-D setting as the one shown previously in Fig. 9 that was obtained with only two objectives. An

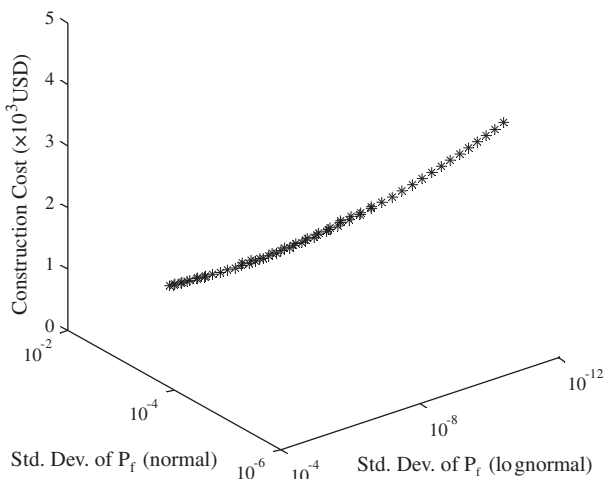


Fig. 12. The Pareto Front at convergence obtained by NSGA-II based on three objectives.

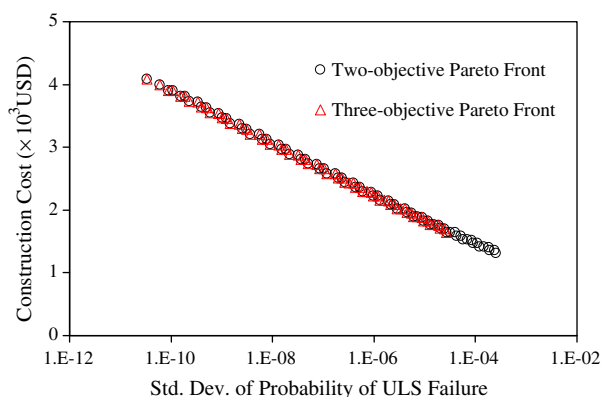


Fig. 13. Comparison of the Pareto Fronts obtained based on two objectives versus three objectives.

interesting observation is in order. When the cost is greater than 1647 USD, the trade-off relationship between cost and robustness is practically the same for the two Pareto Fronts. Thus, for designs with cost greater than 1647 USD, they are insensitive to the assumed distribution type (normal or lognormal). However, the least cost design ($B = 2.4 \text{ m}$, $L = 2.4 \text{ m}$) optimized for three objectives costs 1647.6 USD, which is higher than the cost of the least cost design ($B = 2.1 \text{ m}$, $L = 2.1 \text{ m}$) obtained previously with two-objective optimization. This implies that the design that is insensitive to the assumed distribution type tend to cost more in the reliability-based design of spread foundations. Fig. 13 also indicates that with an additional objective in the optimization, the number of points in the Pareto Front is reduced. This trend is likely to hold if additional objectives are added.

9. Conclusions

Robustness as one of the design objectives has been illustrated in this paper. In fact, the concept of robustness is incorporated into the reliability-based design to deal with the uncertainty in the estimated sample statistics of soil parameters, which is often a major problem in a reliability-based design. The significance of design robustness has been demonstrated with a reliability-based design of spread foundations.

As demonstrated in this study, there often exists no single best design when multiple design objectives are imposed. Thus, an optimal set of designs, called Pareto Front, in which no design is inferior to others with respect to all objectives, is the best of what an engineer can obtain under such a scenario. The Pareto Front specifies a trade-off relationship between cost and robustness, after satisfying all safety requirements. This trade-off relationship enables the engineer to make a more informed design decision. Finally, it has been demonstrated through the example design of spread foundations that NSGA-II is an effective and efficient tool for performing multi-objective optimization for establishing a Pareto Front.

It should be noted that the robust geotechnical design (RGD) is not a methodology to compete with the traditional design approaches; rather, it is a complementary design strategy. With the RGD approach, the focus is to satisfy three design objectives, namely safety, cost, and robustness. In this paper, safety is implemented as a constraint in the form of reliability, and thus the focus of the paper is the reliability-based RGD approach. The RGD methodology is expected to work equally well in a factor-of-safety-based setting. Finally, robustness, which is usually not considered explicitly in a traditional geotechnical design, may have to be achieved at a higher cost, as demonstrated in the example study. Thus, the development of Pareto Front is necessary for trade-off consideration in a robust design. This paper is the first step in developing the RGD methodology. This methodology is being adapted and refined in an ongoing research on design of supported excavation system at Clemson University, which involves challenging numerical and simulation processes within the RGD framework. Further investigations by interested parties are encouraged to advance this design methodology.

Acknowledgments

The study on which this paper is based has been supported in part by National Science Foundation through Grant CMMI-1200117 ("Transforming Robust Design Concept into a Novel Geotechnical Design Tool") and the Glenn Department of Civil Engineering, Clemson University. The results and opinions expressed in this paper do not necessarily reflect the view and policies of the National Science Foundation.

References

- [1] Whitman RV. Organizing and evaluating uncertainty in geotechnical engineering. *J Geotech Geoenviron Eng* 2000;126(7):583–93.
- [2] Casagrande A. The role of the “calculated risk” in earthwork and foundation engineering. *J Soil Mech Found Div* 1965;91(4):1–40.
- [3] Whitman RV. Evaluating calculated risk in geotechnical engineering. *J Geotech Eng* 1984;110(2):143–88.
- [4] Ang AHS, Tang WH. Probability concepts in engineering planning and design: decision, risk, and reliability, vol. 2. New York: Wiley; 1984.
- [5] Harr ME. Reliability-based design in civil engineering. New York: McGraw-Hill; 1987.
- [6] Wu TH, Tang WH, Sangrey DA, Baecher GB. Reliability of offshore foundations—state-of-the-art. *J Geotech Eng* 1989;115(2):157–78.
- [7] Christian JT, Ladd CC, Baecher GB. Reliability applied to slope stability analysis. *J Geotech Eng* 1994;120(12):2180–207.
- [8] Baecher GB, Christian JT. Reliability and statistics in geotechnical engineering. New York: Wiley; 2003.
- [9] Phoon KK, Kulhawy FH, Grigoriu MD. Development of a reliability-based design framework for transmission line structure foundations. *J Geotech Geoenviron Eng* 2003;129(9):798–806.
- [10] Phoon KK, Kulhawy FH, Grigoriu MD. Multiple resistance factor design for shallow transmission line structure foundations. *J Geotech Geoenviron Eng* 2003;129(9):807–18.
- [11] Fenton GA, Griffiths DV. Risk assessment in geotechnical engineering. New York: Wiley; 2008.
- [12] Najjar SS, Gilbert RB. Importance of lower-bound capacities in the design of deep foundations. *J Geotech Geoenviron Eng* 2009;135(7):890–900.
- [13] Wang Y. Reliability-based economic design optimization of spread foundations. *J Geotech Geoenviron Eng* 2009;135(7):954–9.
- [14] Wang Y. Reliability-based design of spread foundations by Monte Carlo Simulations. *Géotechnique* 2011;61(8):677–85.
- [15] Zhang J, Zhang LM, Tang WH. Reliability-based optimization of geotechnical systems. *J Geotech Geoenviron Eng* 2011;137(12):1211–21.
- [16] Phoon KK, Kulhawy FH. Characterization of geotechnical variability. *Can Geotech J* 1999;36(4):612–24.
- [17] Taguchi G. Introduction to quality engineering: designing quality into products and processes. New York: Quality Resources; 1986.
- [18] Chen W, Allen JK, Mistree F, Tsui KL. A procedure for robust design: minimizing variations caused by noise factors and control factors. *J Mech Des* 1996;118(4):478–85.
- [19] Lee KH, Park GJ. Robust optimization considering tolerances of design variables. *Comput Struct* 2001;79(1):77–86.
- [20] Kang Z. Robust design optimization of structures under uncertainties. Aachen, Germany: Shaker Verlag; 2005.
- [21] Kumar A, Nair PB, Keane AJ, Shahpar S. Robust design using Bayesian Monte Carlo. *Int J Numer Methods Eng* 2008(73):1497–517.
- [22] Lagaros ND, Vagelis P, Papadrakakis M. Neurocomputing strategies for solving reliability-robust design optimization problems. *Eng Comput* 2010;27(7):819–40.
- [23] Orr TLL. Design examples for the Eurocode 7 Workshop. Proceedings of the international workshop on the evaluation of Eurocode, vol. 7. Dublin: Trinity College; 2005. p. 67–74.
- [24] Orr TLL, Breyse D. Eurocode 7 and reliability-based design. In: Phoon KK, editor. Reliability-based design in geotechnical engineering: computations and applications. London: Taylor & Francis; 2008. p. 298–343 [chapter 8].
- [25] Akbas SO. Deterministic and probabilistic assessment of settlements of shallow foundations in cohesionless soils. Ph.D. thesis, Cornell Univ., Ithaca, New York; 2007.
- [26] Orr TLL, Farrell ER. Geotechnical design to Eurocode 7. Berlin: Springer; 1999.
- [27] Craig RF. Craig's soil mechanics. London: Taylor & Francis; 2004.
- [28] Poulos HG, Davis EH. Elastic solutions for soil and rock mechanics. New York: Wiley; 1974.
- [29] Wang Y, Kulhawy FH. Economic design optimization of foundations. *J Geotech Geoenviron Eng* 2008;134(8):1097–105.
- [30] Phadke MS. Quality engineering using robust design. Englewood Cliffs, New Jersey: Prentice Hall; 1989.
- [31] Beyer H, Sendhoff B. Robust optimization—a comprehensive survey. *Comput Method Appl M* 2007;196(33–34):3190–218.
- [32] Park GJ, Lee TH, Lee K, Hwang KH. Robust design: an overview. *AIAA J* 2006;44(1):181–91.
- [33] Ait Brik B, Ghanmi S, Bouhaddi N, Cogan S. Robust design in structural mechanics. *Int J Comput Methods Eng Sci Mech* 2007;8(1):39–49.
- [34] Papadopoulos V, Lagaros ND. Vulnerability-based robust design optimization of imperfect shell structures. *Struc Saf* 2009;31(6):475–82.
- [35] Doltsinis I, Kang Z. Robust design of structures using optimization methods. *Comput Method Appl M* 2004;193(23–26):2221–37.
- [36] Doltsinis I, Kang Z. Robust design of non-linear structures using optimization methods. *Comput Method Appl M* 2005;194(12–16):1779–95.
- [37] Hasofer AM, Lind NC. Exact and invariant second-moment code format. *J Eng Mech Div* 1974;100(1):111–21.
- [38] Zhao YG, Ono T. New point estimates for probability moments. *J Eng Mech* 2000;126(4):433–6.
- [39] Zhao YG, Ono T. Moment methods for structural reliability. *Struc Saf* 2001;23(1):47–75.
- [40] Hohenbichler M, Rackwitz R. Non-normal dependent vectors in structural safety. *J Eng Mech Div* 1981;107(6):1227–38.
- [41] Deb K, Pratap A, Agarwal S, Meyarivan T. A fast and elitist multiobjective genetic algorithm: NSGA-II. *IEEE Trans Evol Comput* 2002;6(2):182–97.
- [42] Gencturk B, Elnashai AS. Multi-objective optimal seismic design of buildings using advanced engineering materials. Research Report 11-01, Department of Civil and Environmental Engineering, University of Illinois at Urbana-Champaign: Mid-America Earthquake Center; 2011.
- [43] Jimenez R, Sitar N. The importance of distribution types on finite element analyses of foundation settlement. *Comput Geotech* 2009;36(3):474–83.
- [44] Wen YK, Kang YJ. Minimum building life-cycle cost design criteria. II: applications. *J Struct Eng* 2001;127(3):338–46.
- [45] Ang AHS. Life-cycle considerations in risk-informed decisions for design of civil infrastructures. *Struct Infrastruct E* 2011;7(1–2):3–9.
- [46] Parkinson A, Sorensen C, Pourhassan N. A general approach for robust optimal design. *J Mech Des* 1993;115(1):74–80.